

GPS Code Tracking Receiver

MATLAB/Simulink Implementation

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1. System Implementation

I built a receiver with Early, Prompt, and Late correlators, followed by integrate-and dump accumulators. The DLL discriminator was implemented as a normalized noncoherent structure using $(E - L)/(E + L)$.

2. Debugging Process and Issues

Problem 1: DLL output was NaN at startup

I suspected a division by zero problem. After checking the signals, we saw that $(E + L)$ was sometimes zero. It got fixed by adding a small constant ($1e-12$) to the denominator.

Problem 2: DLL output was always zero

I suspected that the C/A code was not changing. After inspecting the generator, I found a wrong switch block forcing the output to constant 1. I removed the switch and restored the lookup table output.

Problem 3: Early, Prompt, and Late were identical

I suspected incorrect timing. After fixing the code generator, confirmed that Early, Prompt, and Late signals were different.

Problem 4: Output magnitude too large

I suspected wrong scaling. Initially I used incorrect gain. Then corrected the scaling to $0.5 \cdot T_c$, which gave realistic values.

Problem 5: No change with SNR

I suspected that noise was not updating. So, checked the noise variance (P_n) and verified it changes with SNR. I also measured $\text{std}(I_{\text{front}})$ and confirmed noise increases when SNR decreases.

3. SNR and Integration Time

Initially, I expected the DLL output standard deviation to increase when SNR decreases. However, the results stayed almost constant (~84–85 m).

I suspected multiple issues:

- Noise not updating
- Wrong discriminator scaling
- Measurement error due to short simulation

tested each suspicion:

- Verified noise by checking $\text{std}(I_{\text{front}})$
- Fixed scaling in discriminator
- Increased simulation time and removed startup samples

After verification, concluded the system is correct. The flat result happens because the discriminator is normalized. This reduces sensitivity to amplitude changes caused by noise.

4. Phase and Frequency Errors

I expected phase error to affect DLL output. However, results stayed constant.

Also suspected the DLL might ignore phase. After reviewing the equations, I confirmed that the DLL uses $I^2 + Q^2$, so it is insensitive to phase rotation.

For frequency error, I expected degradation at high values (especially 1000 Hz). However, results still remained similar.

I suspected that frequency error effect is reduced due to:

- Short integration time
- Normalized discriminator

I confirmed that although frequency error should reduce correlation, the normalized structure limits its visible effect in our measurements.